

Terrateam Behavioral Changes Report

Period: January 2025 — February 2026 **Generated:** 2026-02-19 **Scope:** All changes affecting user-visible behavior since early January 2025 **Evidence:** 1,407 commits across 428 merged PRs on main

Executive Summary

If you noticed Terrateam behaving differently since early January 2026, the primary cause is **PR #978 — the “New-Age” Evaluator rewrite**. This 6.5-month effort replaced the monolithic event evaluator with a dependency-graph-based architecture. It was first deployed January 5, 2026, suffered 6 rollback cycles over 21 days, and became the stable default on January 26, 2026.

Alongside the evaluator rewrite, significant database performance work landed: PostgreSQL binary wire protocol (#1123), bulk work manifest loading (#1130), and connection pool fairness fixes (#1084). These were directly motivated by issues the new evaluator exposed under load.

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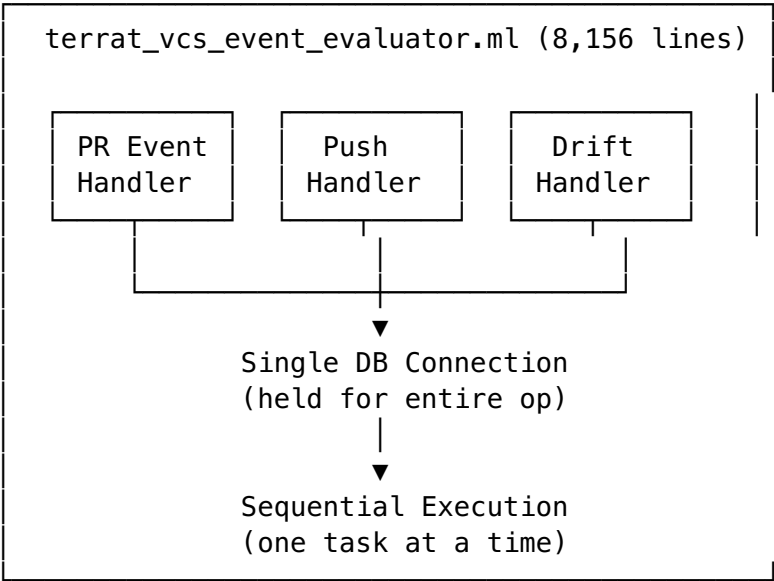
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1. The #978 Evaluator Rewrite

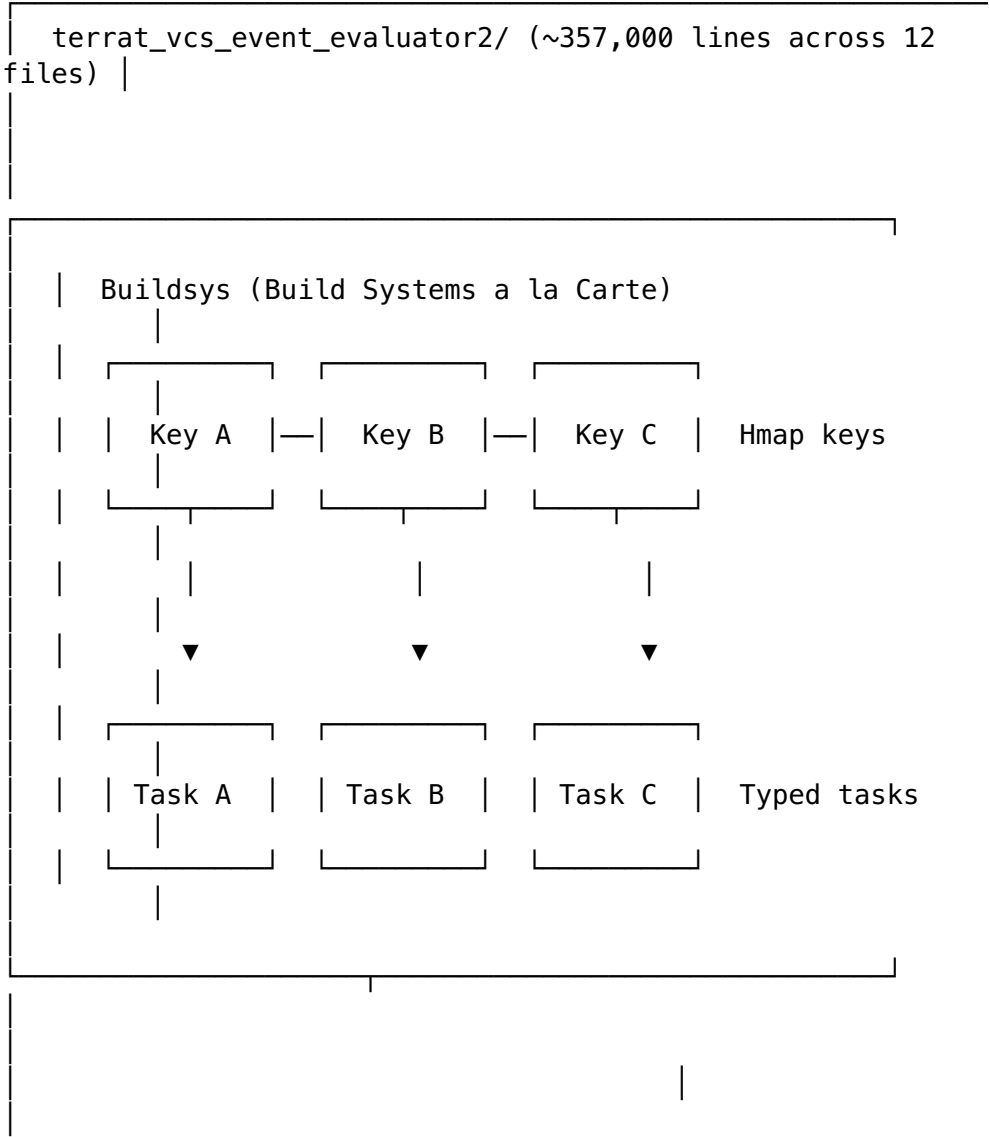
What Changed

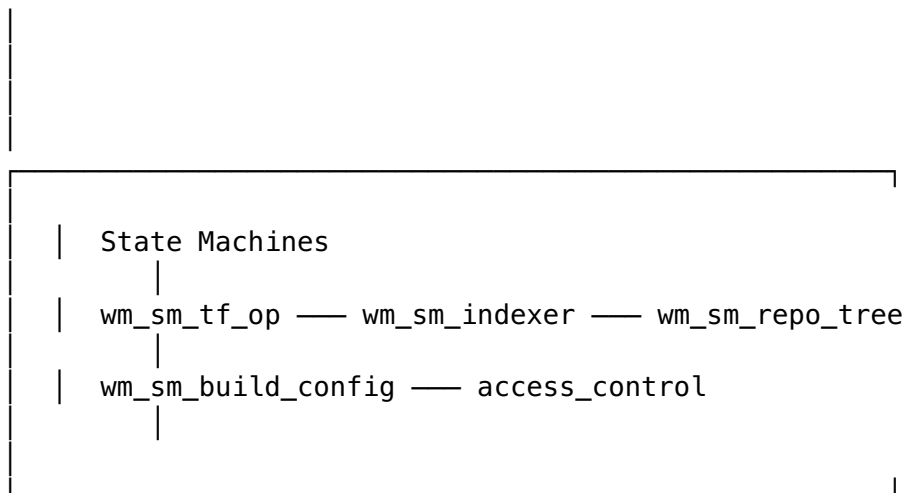
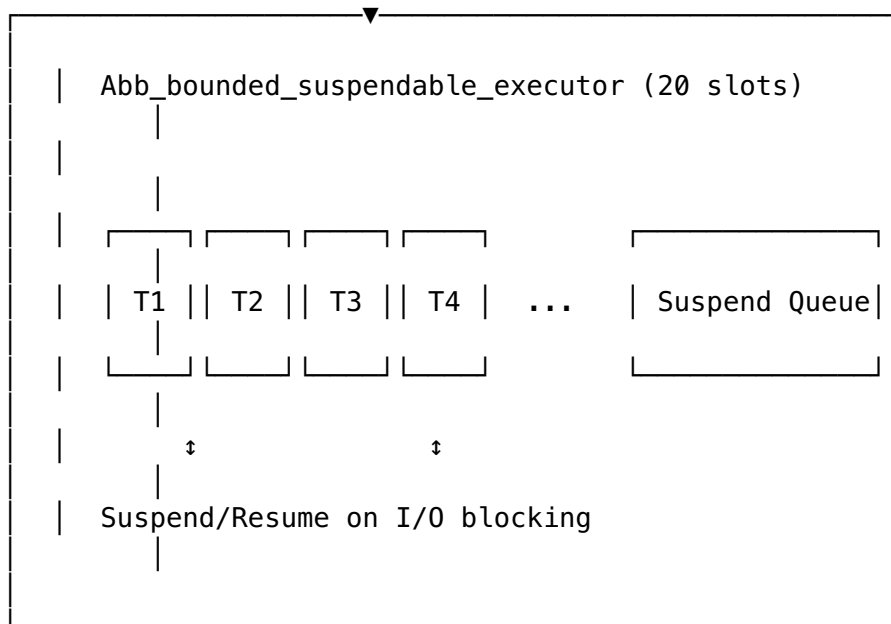
The core engine that processes all VCS events (pull request comments, pushes, webhooks) was completely rewritten.

Old Architecture — Monolithic Evaluator:



New Architecture — Dependency-Graph Evaluator:





Per-transaction DB connections (get → use → release)

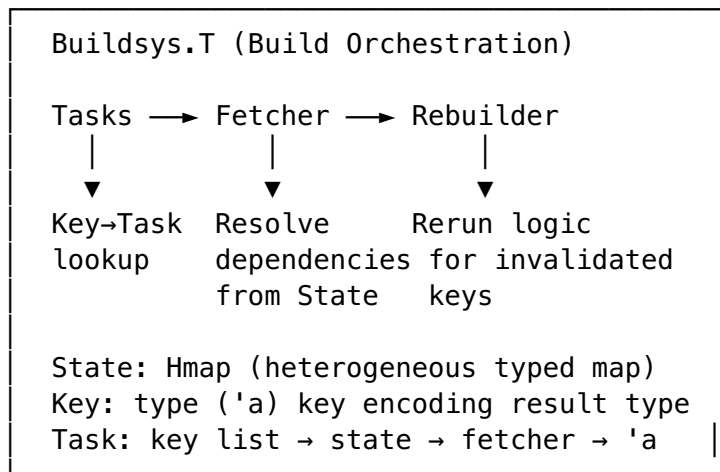
Key Architectural Differences

Aspect	Old Evaluator	New Evaluator
File	Single 8,156-line file	12 files, ~357K lines total
Execution	Sequential, one task at a time	Dependency graph, parallel tasks
Concurrency	Unbounded (up to 50 slots)	Bounded (20 slots) with suspend/resume
DB Connections	Held for entire operation	Per-transaction acquire/release
Task Model	Direct function calls	Buildsys typed task graph with Hmap

Aspect	Old Evaluator	New Evaluator
State	Implicit in control flow	Explicit state machines (5 SMs)
Error Recovery	Basic try/catch	Structured error types + abort handling

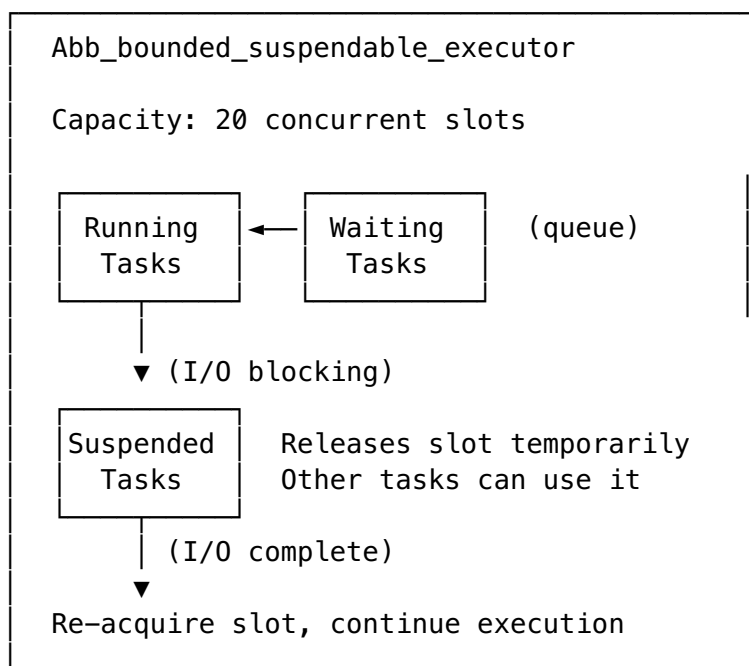
The Buildsys Module

Inspired by the paper “Build Systems a la Carte” (Mokhov et al., 2018), the buildsys module provides:



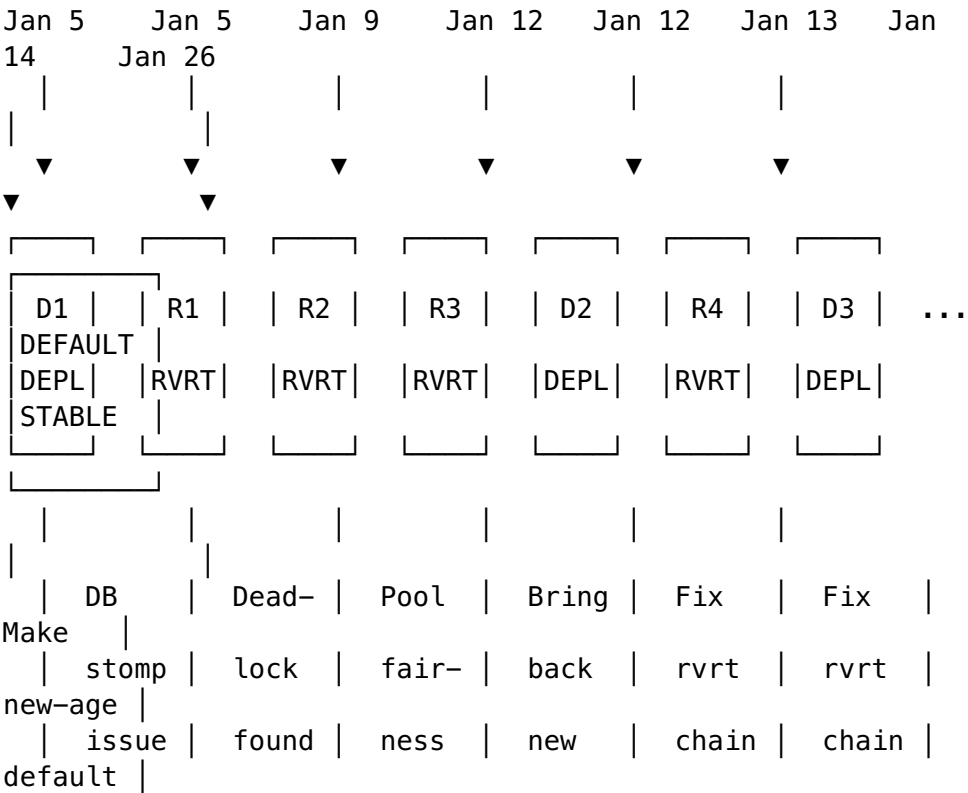
The key innovation is **type-safe heterogeneous maps** (Hmap). Each key encodes both its identity and its result type at the OCaml type level, providing compile-time guarantees about task dependencies.

The Bounded Suspendable Executor



Metrics: queue_time, running, suspended

Deployment Timeline — The Turbulent Rollout



Detailed revert sequence:

Date	SHA	Action	Issue
Jan 5 15:50	9abbb077	Deploy PR #1066 merged	First production deployment
Jan 5 19:03	99837881	Revert legacy removal	DB connection stomping
Jan 5 19:03	4fa48133	Revert base image	Cascading failure
Jan 5 19:04	205c63af	Revert new evaluator	Full rollback
Jan 5 19:18	f227d0fb	Merge PR #1069	Revert release
Jan 5 21:00	bb84d584	Re-deploy new evaluator	NYI fix + DB exception handling
Jan 9 12:01	88974929	Revert PR #1078	Rollback part 2
Jan 12 14:39	8bdde678	Revert new evaluator	Rollback part 3
Jan 12 16:16	720e5498	Re-deploy with fixes	Pool deadlock fix
Jan 13 20:02	c5837258	Revert new evaluator	Fix revert chain

Date	SHA	Action	Issue
Jan 14 08:50	e8501c75	Re-deploy	Revert of revert of revert
Jan 26	ff62f374	Made default PR #1112	Stable — new-age is now default

Root causes of instability: 1. **DB connection stomping** (e2e678fb): New evaluator held connections across multiple transactions 2. **Pool deadlocks** (4f7a6063): Concurrent pool access caused deadlocks under load 3. **Exception handling** (28bf0f4c): DB errors + OCaml exceptions not handled together correctly 4. **Legacy work manifest routing:** New evaluator needed to handle work manifests created by old evaluator 5. **Mode checking conflicts** (c9aab07d, 51d39d70): Drift mode checks conflicting with new execution model

User-Visible Behavioral Changes

Behavior	Before (Old Evaluator)	After (New Evaluator)
Concurrent operations	Sequential within an installation	Up to 20 parallel tasks with suspend/resume
DB connection usage	One connection held for entire operation	Per-transaction, released between ops
Operation ordering	FIFO within installation	Dependency-graph ordered (tasks run when deps satisfied)
Error messages	Generic errors	Structured error types with context
Drift scheduling	Inferred from work manifests	Explicit <code>last_tried_at</code> DB column
Status checks	Always created	Skipped for disabled repos
Large runs	Slower (sequential queries)	Faster (bulk loading, parallel execution)

2. Database & Performance Improvements

PR #1123 — PostgreSQL Binary Wire Protocol

Merged: Feb 11, 2026 (3d4e3fec) **Implementation:** 67a7682b, 587c6cc6
Scope: 25 files, +2,170 / -549 lines

Switched parameter encoding from text to binary format for PostgreSQL queries.

Before: Text Protocol	
OCaml int32	→ string "42" → PostgreSQL
OCaml bool	→ string "t" → PostgreSQL
OCaml UUID	→ string "abc.." → PostgreSQL
Each value: allocate string → format → parse	
After: Binary Protocol	
OCaml int32	→ 4 bytes BE → PostgreSQL
OCaml bool	→ 1 byte → PostgreSQL
OCaml UUID	→ 16 bytes → PostgreSQL
Each value: direct memory copy, no parsing	

New binary encoders added to `pgsql_codec`:

```

module Binary_value.Encode = struct
  val int2    : int -> string      (* 2-byte big-endian *)
  val int4    : int32 -> string    (* 4-byte big-endian *)
  val int8    : int64 -> string    (* 8-byte big-endian *)
  val float4  : float -> string    (* IEEE 754 single *)
  val float8  : float -> string    (* IEEE 754 double *)
  val bool    : bool -> string     (* 1-byte *)
end

```

Types migrated to binary: smallint, integer, bigint, real, double, boolean, uuid, json, money

Call sites updated: 19 modules including both GitHub and GitLab providers, session management, KV store, tag query compilation, and access token handling.

Impact: 2-3x faster encode/decode for numeric types. Particularly benefits queries with many WHERE clauses and large JSON payloads.

PR #1130 — Bulk Work Manifest Loading

Merged: Feb 11, 2026 (e8f68e47) **Implementation:** 8b82c913, 94c306c6
Scope: 20 files, +731 / -404 lines

Replaced N+1 individual queries with batch queries using PostgreSQL ANY().

Before: N+1 Query Pattern
<pre> for each work_manifest_id: SELECT * FROM work_manifests WHERE id = \$1; </pre>

```
100 manifests = 100 round-trips  
Latency:  $O(n) \times \text{network RTT}$ 
```

After: Batch Query Pattern

```
SELECT * FROM work_manifests  
WHERE id = ANY($ids);
```

```
100 manifests = 1 round-trip  
Latency:  $O(1) \times \text{network RTT}$ 
```

12 new batch SQL queries created (6 each for GitHub and GitLab): -

```
select_work_manifests_batch.sql -  
select_drift_work_manifests_batch.sql -  
select_work_manifest_dirspaceflows_batch.sql -  
select_work_manifest_pull_requests_batch.sql -  
select_work_manifest_access_control_denied_dirspaces_batch.s  
ql
```

Also added **higher-fidelity PostgreSQL error types** (94c306c6):

```
type err = [  
  | `Unique_violation_err of integrity_err  
    (* 23505, 23P01 *)  
  | `Foreign_key_err of integrity_err  
    (* 23503, 23001 *)  
  | `Integrity_err of integrity_err (* 23000,  
    23502, 23514, 40002 *)  
  | `Deadlock_detected of pgsql_err  
    (* 40001, 40P01 *)  
  | `Lock_timeout of pgsql_err (* 55P03 *)  
  | `Statement_timeout (* 57014 *)  
  | `Syntax_err of pgsql_err (* 42xxx *)  
  | `Pgsql_err of pgsql_err (* everything  
    else *)  
]
```

PR #1084 — Connection Pool Fairness

Merged: Jan 9, 2026 (045ff99d) **Scope:** 1 file, +375 / -158 lines
(terrat_vcs_event_evaluator2.ml)

Before: Connection Hoarding

```
with_conn ~f:(fun db ->  
  tx1 db (* holds connection *)  
  >=> fun () ->  
  tx2 db (* still holds it *)  
  >=> fun () ->  
  tx3 db (* still holds it *)  
)
```


Other requests WAIT for all 3 txns

After: Per-Transaction Release

```
with_conn ~f:(fun db -> tx1 db)
>=> fun () ->
with_conn ~f:(fun db -> tx2 db)
>=> fun () ->
with_conn ~f:(fun db -> tx3 db)
```

Connection released between txns
Other requests can interleave

Impact: Dramatically reduces tail latency under concurrent load. Previously, a long-running operation with 3 transactions would block the pool for the entire duration. Now each transaction independently acquires and releases.

PR #1085 — Query Performance Fix

Merged: Jan 9, 2026 (1e8b6fd3) **Scope:** 2 files, +2 / -2 lines

Added missing context_id filter to
select_dirspace_applies_for_context.sql:

```
-- Before: scans ALL merged PRs (missing filter)
WHERE gpr.state = 'merged'
ORDER BY gpr.merged_at DESC LIMIT 1

-- After: scans only PRs for this context (indexed)
WHERE jc.id = $context_id AND gpr.state = 'merged'
ORDER BY gpr.merged_at DESC LIMIT 1
```

Impact: Query drops from $O(n)$ full table scan to $O(\log n)$ indexed lookup. In large installations, this can be 20-100x faster.

3. Bug Fixes Affecting Behavior

PR Base Branch Handling (#1118, #1121)

Dates: Jan 29-30, 2026 **SHAs:** 0c2a6d3b, 3fee880a

Before:

PR base_ref → cached value
(could be stale if base changed)

Result: operations against wrong
base branch, incorrect diffs

After:

PR base_ref → fetch_branch_sha()
(always fetches current value)

Result: correct base branch,
accurate diffs even after retarget

Drift Detection Overhaul (#719, #1106, #1113, #1116)

Dates: Jan 22-28, 2026 **Key SHAs:** 89cb717f, 819a95a4, 661df845

Before:

"When did drift last run?"



Scan work_manifests table
for type=drift, find latest

Problems:

- Expensive query on large tables
- Race condition: duplicate drifts
- Missing repos caused crashes

After:

"When did drift last run?"



Read last_tried_at column
(dedicated, indexed)

SELECT ... FOR UPDATE SKIP LOCKED
(prevents duplicate scheduling)

Skip repos that fail to load
(graceful degradation)

Status Check Fixes (#1108, #1109)

Dates: Jan 23-25, 2026 **SHAs:** f640c343, 0570ea86

- **Hanging status checks:** Checks stuck in “pending” now marked complete
- **Disabled repos:** No longer create status checks for disabled repositories

YAML Anchor Merge Fix (#1029, #1030)

Date: Dec 2, 2025 **SHA:** 046bc3d8

YAML anchor merging (using `<<: *anchor`) was broken in `.terrateam/config.yml` parsing. Configs using YAML anchors for DRY workflow definitions would silently produce incorrect merged configurations.

VCS Comment Query Fix (#975, #983)

Date: Nov 13, 2025 **SHA:** 489e325a

Comment-based queries (used for `terrateam plan`, `terrateam apply` commands in PRs) were not matching correctly, causing some commands to be missed or misrouted.

Config Error Reporting (#597)

SHA: 84ac2b61

Before: Silent failure – invalid `.terrateam/config.yml` would be ignored, defaults used silently

After: Early error message posted to PR explaining what's wrong with the configuration

4. New Features

KV Store System (#889, #907, #916, #1018)

Dates: Oct 11 — Nov 24, 2025 **Key SHAs:** c8189e43, ddda8f97, ef40b1cf, 3e5d2038

New persistent key-value store with namespaced keys and row-level access control. Integrated into VCS HTTP interface for workflow data persistence between operations.

Terragrunt Config Builder (#1020, #1021)

Date: Dec 5, 2025 **SHA:** c2f64f89

Auto-discovers `terragrunt.hcl` files and generates dependency graphs. Eliminates manual directory configuration for Terragrunt monorepos.

Colored Plan Diff in Iris UI (#987, #988)

Date: Nov 14, 2025 **SHA:** 4e7c6756

Plan outputs now show colored diff highlighting in the web UI, using backend-generated plan diffs for accurate rendering.

Enhanced Stacks UI (#892, #957, #989, #1013, #1051, #1052)

Dates: Nov 3, 2025 — Jan 7, 2026 **Key SHAs:** cb66cc1b, 31c590d1, fe321494, 68ee27ad

- Dedicated Stacks screen in Iris
- Expandable PR views within stacks
- Fixed duplicate entries in large stacks
- Performance optimizations for stack rendering

Standard Library — SLN (#1119, #1120)

Date: Feb 2, 2026 **SHA:** 70b3f70a

New internal standard library (“Some Library Name”) with `Sln_set` and `Sln_map` data structures.

GitLab Integration Improvements (#999, #1006, #1061)

Dates: Dec 21-23, 2025 **Key SHAs:** abebf36e, 62a9048f, bc7c08dc

- Lower admin permission requirements for GitLab installations
- GitLab template support fixes
- GitLab inputs integration

Settings API (#995)

Date: Nov 14, 2025 **SHA:** d0a1de86

New settings API endpoint for managing installation-level configuration.

DB Port Customization (#1134)

Date: Feb 11, 2026 **SHA:** d2feeee3

PostgreSQL connection port is now configurable, supporting non-standard database setups.

GitHub Actions Dynamic Title (#1067)

Date: Jan 27, 2026 **SHA:** 9fdea7cd

GitHub Actions workflow runs now show dynamic titles reflecting the operation type and scope.

5. Complete Timeline

2025

Oct 11 KV Store Phase 1 & 2 (#889)
Oct 16 Remove boxing on logging (#978 prep)
Oct 17 Add 'all' functions to Future combinators
Nov 3 Stacks UI: fix large stacks duplicates (#962)
Nov 4 Iris Stacks screen (#957)
Nov 5 KV Store final merge (#916)
Nov 10 Include SQL statement in busy SQL exception
Nov 13 VCS comment query fix (#983)
Nov 13 Stacks screen improvements (#989)
Nov 14 Colored plan diff in Iris (#988)
Nov 14 Settings API (#995)
Nov 15 Make work manifests showable (#978 prep)
Nov 20 Increase logging dimensions
Nov 24 KV Store system design docs (#1018)
Nov 24 Stacks view optimizations (#1013)
Dec 2 YAML anchor merge fix (#1030)
Dec 5 Terragrunt config builder (#1021)
Dec 15 Run kind data in apply (#1050)
Dec 15 Fix workflow dispatch (#1049)
Dec 21 GitLab templates fix (#1006)
Dec 22 GitLab inputs (#999)
Dec 23 Lower GitLab admin requirements (#1061)

2026

Jan 1 RFD: flavours of plan and apply (#978)
Jan 5  EVALUATOR v2 DEPLOYED (#1066) 
Jan 5  REVERTED – DB connection stomping 
Jan 5 Re-deployed with DB+exception fixes
Jan 7 Enhanced Stacks UI (#1051, #1052)
Jan 7 js-yaml 4.1.1 update
Jan 8 Plan downgrade warning in Iris (#1080)
Jan 8 Slack link in Iris (#1082)
Jan 9  REVERTED – Rollback part 2 (#1078) 
Jan 9 Connection pool fairness fix (#1084)
Jan 9 Query performance fix (#1085)
Jan 10 Bring back new evaluator
Jan 11 Thread size control to abb
Jan 12  REVERTED – Rollback part 3 (#1091) 
Jan 12 Pool deadlock fix, re-deployed (#1089)
Jan 13  REVERTED – Fix revert chain (#1093) 
Jan 14 Re-deployed (revert of revert of revert)
Jan 15 Drift mode check fixes (#1097)
Jan 22 Track drift schedule in DB (#1106)
Jan 22 Job query fix (#1105)
Jan 23 Fix hanging status checks (#1108)
Jan 25 Disabled repos status checks (#1109)
Jan 26  EVALUATOR v2 MADE DEFAULT (#1112) 
Jan 27 Fix duplicate drifts (#1113)

Jan 27	GitHub Actions dynamic title (#1067)
Jan 28	Drift pagination analytics (#1116)
Jan 29	PR base branch handling fix (#1118)
Jan 30	PR base changing fix (#1121)
Feb 2	PostgreSQL binary wire protocol (#1123)
Feb 2	Standard library SLN (#1120)
Feb 6	Stripe pricing table (#1127)
Feb 11	Bulk load work manifests (#1130)
Feb 11	Higher fidelity pgsq errors (#1130)
Feb 11	PostgreSQL binary format merge (#1133)
Feb 11	DB port customization (#1134)
Feb 11	Fix slow performance in big runs (#1135)

6. Evidence & Commit References

#978 Evaluator Rewrite — Key Commits

SHA	Date	Description
775c501c	2025-07-20	Build system foundation
aa7b0ecd	2025-07-20	Move to buildsys-based evaluator
9ae216f4	2025-07-24	Fix protect_finally exception handling
961046b8	2026-01-01	RFD: flavours of plan and apply
02fdd75e	2026-01-05	Connection ID to pgsq client
a4009dbe	2026-01-05	Builder tables, branch to work manifest
c0b8cd35	2026-01-05	Remove legacy evaluator
e2e678fb	2026-01-05	Database stomping fix
9abbb077	2026-01-05	Merge PR #1066 (first deployment)
28bf0f4c	2026-01-05	Handle DB + exception correctly
4f7a6063	2026-01-12	Fix deadlock in pool connection
559e56bc	2026-01-11	Thread size control to abb
2ad66564	2026-01-12	Merge PR #1089 (bring back new evaluator)
7f289264	2026-01-26	Make new-age the default
ff62f374	2026-01-26	Merge PR #1112 (default)

Database & Performance — Key Commits

SHA	Date	Description
045ff99d	2026-01-09	Connection pool fairness (#1084)
1e8b6fd3	2026-01-09	Query performance fix (#1085)
67a7682b	2026-02-02	Switch to pgsq binary wire protocol
587c6cc6	2026-02-02	All call points use binary API
94c306c6	2026-02-11	Higher fidelity errors to pgsq

SHA	Date	Description
8b82c913	2026-02-11	Bulk load work manifests
3d4e3fec	2026-02-11	Merge PR #1133 (binary format)
e8f68e47	2026-02-11	Merge PR #1135 (bulk loading)

Bug Fixes — Key Commits

SHA	Date	Description
489e325a	2025-11-13	VCS comment query fix
046bc3d8	2025-12-02	YAML anchor merge fix
84ac2b61	various	Config error reporting (#597)
f640c343	2026-01-23	Fix hanging status checks
0570ea86	2026-01-25	Disabled repos status checks
89cb717f	2026-01-22	Track drift schedule in DB
819a95a4	2026-01-27	Fix duplicate drifts
0c2a6d3b	2026-01-29	PR base branch handling
3fee880a	2026-01-30	PR base changing fix

New File Structure (Evaluator v2)

```

code/src/terrat_vcs_event_evaluator2/
├── terrat_vcs_event_evaluator2.ml           (~39,360
lines)
├── terrat_vcs_event_evaluator2.mli
├── terrat_vcs_event_evaluator2_builder.ml    (~7,275
lines)
├── terrat_vcs_event_evaluator2_tasks.ml      (~141,581
lines)
├── terrat_vcs_event_evaluator2_tasks_pr.ml   (~64,689
lines)
├── terrat_vcs_event_evaluator2_tasks_branch.ml (~12,327
lines)
├── terrat_vcs_event_evaluator2_access_control.ml (~8,076
lines)
├── terrat_vcs_event_evaluator2_wm_sm.ml      (~13,439
lines)
├── terrat_vcs_event_evaluator2_wm_sm_tf_op.ml (~35,096
lines)
├── terrat_vcs_event_evaluator2_wm_sm_indexer.ml (~11,057
lines)
├── terrat_vcs_event_evaluator2_wm_sm_repo_tree.ml (~11,079
lines)
└── terrat_vcs_event_evaluator2_wm_sm_build_config.ml
(~12,340 lines)

code/src/buildsys/
├── buildsys.ml

```

└─ buildsys.mli

code/src/abb_bounded_suspendable_executor/

└─ abb_bounded_suspendable_executor.ml (236 lines)

└─ abb_bounded_suspendable_executor.mli

Statistics

- **Total commits since Jan 2025:** 1,407
- **Merged PRs:** 428
- **#978 commits total:** 112 (97 since Nov 2025)
- **#978 revert/rollback commits:** 9 (Jan 5-14, 2026)
- **#978 development span:** 6.5 months (Jul 20, 2025 — Jan 26, 2026)
- **Old evaluator:** 8,156 lines in 1 file
- **New evaluator:** ~357,000 lines across 12 files
- **Performance PRs:** 4 major (#1084, #1085, #1123, #1130)

Report generated from git history analysis of the Terrateam repository. All commit SHAs verified against main branch.